



Review

The impact of biochar on soil carbon sequestration: Meta-analytical approach to evaluating environmental and economic advantages



Supriya Majumder^a, Surama Neogi^a, Tanushree Dutta^a, Michael A. Powel^b, Pabitra Banik^{a,*}

^a Agricultural and Ecological Research Unit, Indian Statistical Institute, Kolkata 700108, India

^b Dept. of Renewable Resources, Faculty of Agriculture, Life and Environmental Sciences (ALES), University of Alberta, Edmonton, CA, Canada

ARTICLE INFO

Keywords:

Biochar
Carbon storage
Crop residues
Greenhouse gas emission
Sequestration
Meta-analysis

ABSTRACT

Soil carbon (SC) is important for food security, ecosystem functioning, and environmental health, especially in light of global climate change. The physico-chemical character of biochar (pyrolyzed crop residue) has been shown to augment SC levels. This review systematically compares the environmental and economic benefits of applying crop residue versus biochar produced from crop residues to soils and the potential implications for SC sequestration. Crop residues enhance the mineralization rate of SC, while biochar can increase or decrease SC depending on the types of biochar/soil and duration. Therefore, converting crop residues to biochar may be more efficient for sequestering SC, but may/may not be more cost-effective. In this review, special emphasis is given to understanding the underlying mechanisms and biogeochemical processes of biochar production, in particular: surface (crystallinity), redox, and ability to control electron transfer reactions. By using meta-analyses, we determined the role of biochar compared to crop residue to enhance the status of organic SC.

1. Introduction

Total C in terrestrial ecosystems is approximately 3170 GT and nearly 80% (2500 GT) of it is found in soil. Soil carbon can be either organic (1550 GT) or inorganic carbon (950 GT). Soil can sequester an additional 10–60 mg C/hm² (1 hm² = 10⁴ m²) (Stout et al., 2016). Soil capture of atmospheric CO₂ produced by fossil fuel utilization is a desirable strategy which will decrease atmospheric concentrations of CO₂ (Olson et al., 2014), mitigate the impact of climate change (Mao et al., 2012), and improve global food security by increasing the rate of crop growth and enhancing soil productivity (Baronti et al., 2010; Shanthi et al., 2013). In developing countries, a 1 Mgha⁻¹yr⁻¹ increase in the soil organic pool could increase their crop production by 32 Mtyr⁻¹ (Lal, 2006). Principal strategies for C sequestration include: i) the restoration of degraded soils through conversion to agricultural land use, ii) the adoption of recommended management practices such as no-till farming; manure and compost-based organic farming, and iii) the use of biochar as a soil amendment (Lal, 2008). According to recent estimates, biochar's potential for net removal of greenhouse gas (GHG) from the atmosphere is approximately 1–1.8 Pg CO₂-C equivalent yr⁻¹ (Fig. 1, from (Paustian et al., 2016; Smith, 2016; Woolf et al., 2010).

Biochar consists of a stable C compound produced as a byproduct of pyrolysis and gasification (Beesley et al., 2011; Lehmann and Joseph, 2015; Spokas et al., 2011a). Biochar is available either as a powder or in

pelletized form. Char is ubiquitous in natural soils, comprising as much as 35% of total organic C (TOC) in fire-impacted soils (Skjemstad et al., 2002). Soil amendment by char is well recognized in archeological sites including the Amazonian Terra Preta (Lehmann and Joseph, 2015). The addition of biochar was found to significantly enhance soil fertility with the increase in soil organic carbon (SOC) and nutrients (Coomes and Miltner, 2016). Soil organic carbon (SOC) is composed of a complex matrix with partially decomposed substances including polysaccharides, lignin, aliphatic biopolymers, tannins, lipids, proteins, and amino sugars derived from plant litter as well as from faunal and microbial biomass (Stockmann et al., 2013). Depending upon the rate of decomposition in soil, the SOC has been classified into active, slow, and inert pools (Phukan and Ramakrishna, 2014). Recently, the potential of biochar in the restoration of degraded lands (e.g., saline soils and mine tailings) has been confirmed (Drake et al., 2016; Muñoz et al., 2016).

Biochar contributes largely to the stable SOC pool while fresh crop residues add to the more bio-accessible fraction of the soil C reserve. This is because the decomposition of biochar proceeds typically at 10–100 times slower than that of uncharred biomass, e.g., fresh crop residues (Paustian et al., 2016). Crop residues left in agricultural fields, on the other hand, can be used as a SOC storage/sequestration strategy by supporting cover crops; conservation tillage provides benefits including nutrient cycling, control of surface runoff of water, wind erosion, and crop production (Blanco-Canqui and Lal, 2009; Cerdà et al.,

* Corresponding author. Agricultural and Ecological Research Unit, Indian Statistical Institute, 203, B.T. Road, Kolkata 700108, India.

E-mail address: pbanik@isical.ac.in (P. Banik).

<https://doi.org/10.1016/j.jenvman.2019.109466>

Received 5 November 2018; Received in revised form 30 July 2019; Accepted 23 August 2019

Available online 02 September 2019

0301-4797/ © 2019 Elsevier Ltd. All rights reserved.

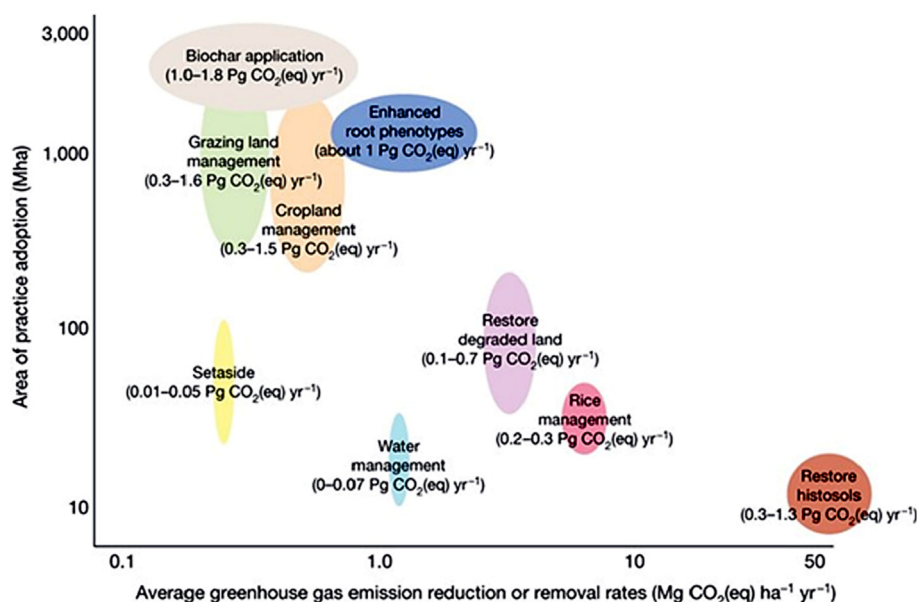


Fig. 1. Global potential for agriculture-based GHG mitigation practices Source: Paustian et al. (2016)). Reprinted by permission from Macmillan Publishers Ltd: [Nature] [Paustian et al. (2016)] Copyright (2016).

2016; Nishigaki et al., 2016). Yazdanpanah (2016) showed that the application of plant residues (e.g., alfalfa) improved the stability and total porosity of soil aggregates depending on the soil type and application rate of amendments. Kenney et al. (2015) found that the removal of residues (e.g., *Zea mays* L. or corn stover) decreased the soil C pool while it increased crop yield. The differing findings in these studies point to the highly variable nature of soils. Since crop residues are sources of organic C within SOC pools, their removal can deplete SOC to a considerable extent (Wegner et al., 2015). Another issue related to crop residue usage is that its collection imposes an additional workload for producers. Although lignocellulosic biomass is a known parent material for SOC (Sposito, 1989), few studies have described crop residues as a C sink as crop residues are traditionally considered a labile C source rather than recalcitrant (Huang et al., 2004; Liu et al., 2014). On the other hand, the application of biochar rich in complex C compounds may enhance or decrease the mineralization rate of soil C, depending on the external factors like type of feedstock and pyrolytic conditions (Zimmerman et al., 2011).

Some of the above-stated impacts of biochar amendment on soil C have been shown to be short-lived as shown in controlled laboratory experiments (Jones et al., 2012). These findings should only be considered in light of short and long-term field observations. Results from long-term field studies are required to calculate rate-dependent contributions from crop residues to SOC pools. This study presents a comprehensive review of work which analyzed the effects of crop residues and biochars on the stable and active SOC pools. The fate of soil C and changes in other physiochemical and biological parameters is also highlighted, with emphasis on particle surface reactions/crystallinity, redox (for biochar), CEC, sorption, electron transfer reactions (Saquing et al., 2016; Xu et al., 2016). The findings are meant to better understand the impact of amending soils with crop residue vs biochar relative to nutrient cycling and availability.

Meta-analysis was used to compare the role of biochar vs crop residue on the status of SOC as long-term methods of soil OC storage.

2. Impact of biochar on soil quality and carbon status

2.1. Properties of biochar: importance of pyrolytic conditions and feedstock type

The physico-chemical properties of biochar are highly contingent

upon the initial biomass feedstock and the pyrolytic conditions including heating rate, reaction medium (including steam), and others (Spokas and Reicosky, 2009). For instance, the nitrogen content of biochar is not substantial due to the radical recombination and relatively low reactivity of N during initial feedstock thermolysis (i.e., biomass), while, phosphorus(P)-rich biochar can be considered a potential renewable P fertilizer. The C: N (Fig. 2) of biochar can be altered to suite various application needs based on biomass type and pyrolytic conditions (especially temperature). The temperature of pyrolysis affects the ratio of products like oil and syngas (H₂ and CO) since bond scission of ligno-cellulosic biomass is temperature dependent. Generally, the characteristics of biochar generated at different heating rates are quite different because fast pyrolysis leads to incomplete carbonization (i.e., unconverted cellulosic and hemicellulosic fractions to biochar) resulting in rapid mineralization. The incomplete carbonization of biochar can be controllable by increasing the residence time during pyrolysis; this will however decrease biochar yield and the C and N content (due to the dehydration of hydroxyl groups and the thermal degradation of ligno-cellulose structures) (Bruun et al., 2012; Hossain et al., 2010). Differences in feedstock and pyrolysis temperature on biochar yield and stability can be assessed by its relative effects on the Brunauer–Emmett–Teller (BET) Theory, ash content, and fixed C versus volatile matter content (Fig. 3). The characterization of recalcitrance C and agronomic performance of 94 different biochars provides evidence regarding the presence of a broad range of volatile versus fixed carbon compounds, a known determinant of biochar stability (Enders et al., 2012). The condensation of aromatic structures and the formation of graphite cores via dehydrogenation can be achieved as the pyrolysis temperature increases, giving it a longer half-life in the soil environment (Shanthi et al., 2013).

Similar to the redox-active quinone moieties of humus (Lovley et al., 1996), biochar is redox active due to the presence of polycondensed aromatic sheets and quinoid functional groups (Chen et al., 2014; Saquing et al., 2016; Xu et al., 2013, 2016). Biochar displays electron shuttling ability over a range of pyrolytic conditions and feedstock types (Klöpffel et al., 2014), which provide versatility for environmental engineering applications that require controlled electron transfer reactions. Using electrochemical, elemental, and spectroscopic studies, Klöpffel et al., 2014 established that chars produced at intermediate to high temperatures (400–700 °C) display the highest capacity to accept and donate electrons and that the pool of redox active moieties in chars

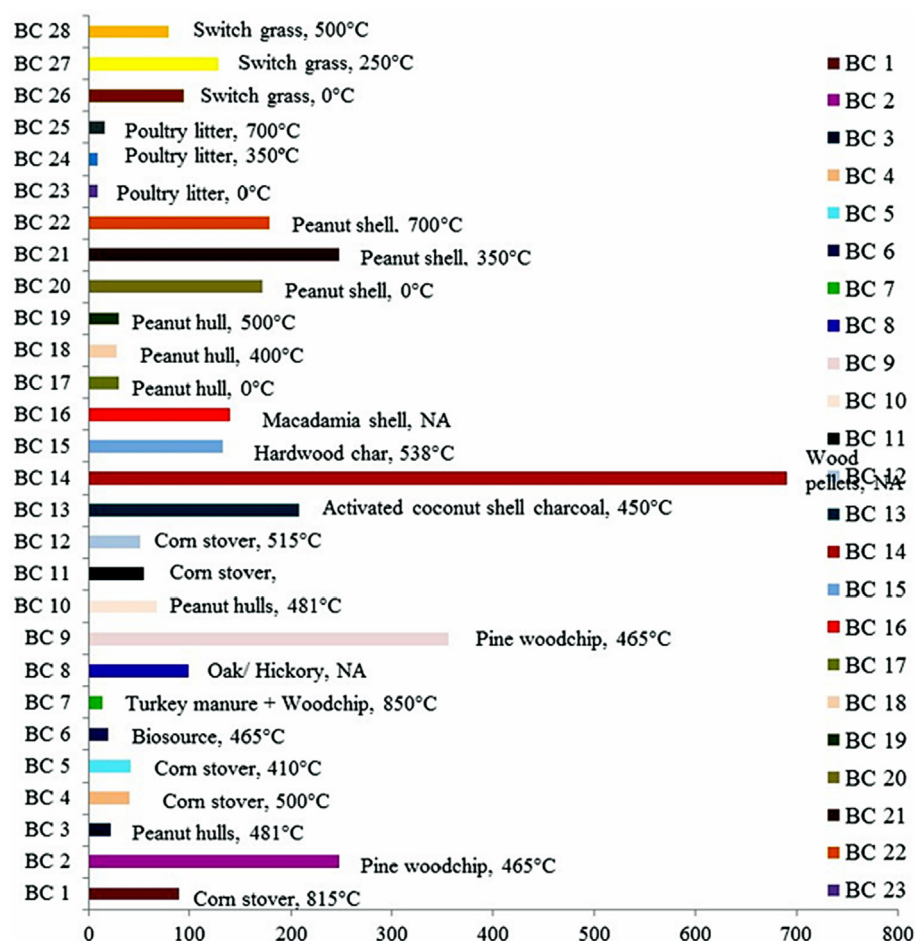


Fig. 2. C/N ratio of biochar prepared from 28 different feedstocks (BC 1 to BC 28) prepared at different pyrolysis temperatures (sources (Novak et al., 2009; Spokas and Reicosky, 2009)).

is dominated by phenols and quinols. In addition, the SOC of biochar participates in microbial signaling to impact chemotaxis (Masiello et al., 2013). The performance of the biochar in soil depends upon its interaction with various components such as the plant roots, microorganisms, SOM, proteins and nutrients dissolved in soil water. The soil-biochar interaction forms organo-mineral-biochar complexes and thus can lead to the modification of the original C matrix of soil. Joseph et al. (2010) and Joseph et al. (2015) provided a detailed overview of the redox characteristics of different biochars and how it influences the electrochemical properties of soils. In summary, biochar mediated redox activity in soil is related to the formation and/or breakdown of organomineral layers during different crop cycles, changes in environmental conditions, its movement through the soil profile, fragmentation of biochar particles, and creating new redox-active surfaces. The redox interactions between soil and biochar are thus of great importance from an environmental point of view and considering the global C budget.

2.2. Impact on SOC

The status of C sequestration using biochar is listed in Table 1a. In most cases, biochars seem to sequester more C than controls with the exception of one study i.e., (Ma et al., 2009). This is likely due to sampling method, e.g., soil depth, or interaction between biochar and soil leading to greater soil C storage via the process of organic matter sorption on biochar surfaces and physical protection (Zimmerman et al., 2011). According to a study by Nguyen et al. (2016), crop residue biochar can stabilize native soil C (negative priming). This can improve the long-term effect of biochar on soil quality with enhanced soil C

storage. This is also partly due to the fact that C in biochar is mainly recalcitrant (Bolan et al., 2012; Mukherjee et al., 2014; Zimmerman, 2010). However, the addition of biochar has been found to exert both positive and negative effects on SOC decomposition, depending on experiment conditions including biochar characteristics, soil type, type of experiment (lab incubation, pot experiment versus field trials), and duration of experiment. As stated earlier, the recalcitrant biochar C varies over a broad range, e.g., the proximate analysis of 94 different biochars (Enders et al., 2012) showed a broad range of volatile versus fixed C contents (volatile matter from 13 to 70% and fixed carbon from 0 to 77%). It was further concluded that a combination of methods may be needed to predict the fate of biochar C in soil.

2.3. Effects on crop yield and soil quality

A review of yield results indicated that the effects of biochar on crop yield is variable (Jeffery et al., 2015). In some cases, biochar increases crop productivity (Baronti et al., 2010; Chan et al., 2008; Qiao-Hong et al., 2014; Shanthi et al., 2013). For example, an observed improvement in the production of cherry tomatoes (by 64% above the controlled soil) can be cited as an example wherein the outputs were attributed to the combined effect of increased nutrient availability (P and N) and soil chemical conditions (Hossain et al., 2010). However, a few studies on the effects of biochar on soil quality and yield indicated that the observed improvements could be short-lived (Quilliam et al., 2012; Spokas et al., 2011a). For example, a three-year field trial of biochar influence on soil quality and plant growth by Jones et al. (2012) show that the addition of biochar caused small and potentially

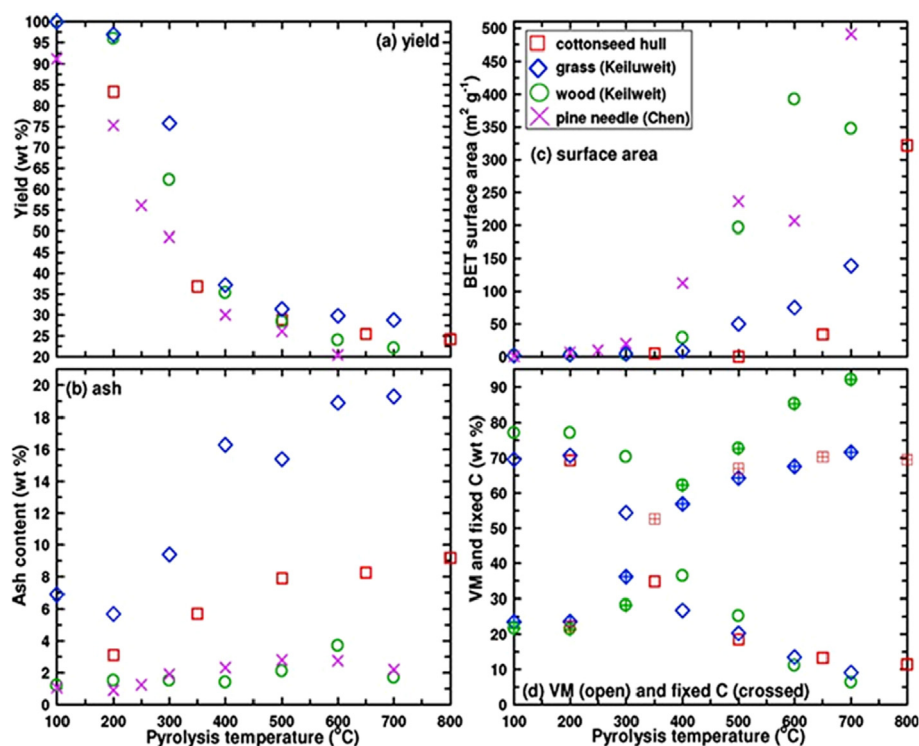


Fig. 3. Comparison of the basic properties of biochars derived from different biomass sources in terms of biochar yield, ash content, volatile matter/fixed C (crossed points), and BET surface area (Uchimiya et al., 2011).

transient changes in temperate agro-ecosystem function. The yield improvements and C cycling resulting from the biochar application were only short-term effects; these contradictory observations could not be verified due to lack of long-term field trials. Liang et al. (2010), suggested that sorption capacity of organic compounds in the biochar might decrease upon aging. Nevertheless, the impact of biochar on long-term yield, availability of N for microorganisms, and for crop uptake is complicated and requires in-depth studies based on individual experimental parameters.

3. Impacts of fresh crop residues on soil quality and carbon status

3.1. Impacts on soil C status

Crop residues refer to any biomass left after the harvest of straw, stubble, husk, and stover, or leaf litter; it is known that the incorporation of crop residues enhances soil C storage (Table 1b). Also, crop residues provide stable soil organic fractions like humic substances. Conservation tillage is practiced to reduce loss of SOC by

oxidative mineralization. Moreover, the time for the addition of residues plays a critical role as a greater fraction of C is sequestered in the early incorporation of crop residues (e.g., 63 days before transplanting) than in late incorporation (14 days before transplanting) (Witt et al., 2000). SOC sequestration is influenced by the extra organic C inputs from crop residues. For example, a significant positive correlation ($R^2 = 0.80$) between cumulative straw C input and SOC sequestered was reported in a meta-analysis of changes in C dynamics due to straw-C inputs (Liu et al., 2014). An additional input of organic C occurs due to higher crop productivity of the amended soils.

3.2. Effects of residue incorporation on the active fraction of C in soil

The application of crop residues can increase the active fraction of C in soil. Recently (Abbasi et al., 2015) showed that the incorporation of plant residues strongly affects the C mineralization-immobilization turnover rate of soils. Increased turnover rates may lead to an increased risk of C mineralization and release CH_4 and CO_2 . This GHG emission can be reduced by practicing conservation tillage to minimize the

Table 1a
Status of C sequestration using crop residue-derived biochars.

Order	Feedstock/Residue type	Crops	Duration of experiment	Soil C status	Reference
1	Wheat straw	Maize	5 months	4.9–6.3 g/kg increase in SOC	Zhang et al. (2011)
2	Wheat straw	Rice paddy	2 years	12.6–12.8 g/kg increase in SOC	Zhang et al. (2012)
3	Wheat straw	Rice	5 months	12.8–13.4 g/kg increase in SOC	Zhang et al. (2010)
4	Green waste	Radish	6 weeks	43 g/kg increase in C content	Chanet al. (2008)
5	Lump charcoal, hickory	–	> 79 days	C sequestered ~ 64% of control	Rogovska et al. (2011)
6	Oak wood	Artificially degraded soil	4 months	0.5% increase in soil carbon	Mukherjee et al. (2014)
7	Rubber tree	Rice	54 days	0.52% increase in SOC	Shanthy et al. (2013)
8	Wood Chip	Maize, Grass	3 years	0.26–0.51% increase in total carbon	Jones et al. (2012)
9	Organic waste substrate	Maize, wheat	5 years	29.0–38.1% SOC derived from maize residue 611.9–73.8% SOC derived from wheat residue	(Dong et al., 2018)
10	Incubation study	–	–	6.4% increase in TOC in 0–3 cm depth	Laird et al. (2010a,b)
11	Incubation study	–	–	18.3–29.2 g/kg increase in SOC (Switch grass)	Novak et al. (2010)
12	Wheat straw	Rapeseed, sweet potato	5 years	76.29% increase in SOC	Zhang et al. (2018)

Table 1b
Status of C sequestration using crop residues.

Order	Feedstock/Residue type	Crops	Duration of experiment	Soil C status	Reference
1	Straw	Maize	10 years	increase in TC (0.56 g/kg), MBC ^a (87 mg/kg), and labile C (0.68 g/kg)	Xu et al. (2011)
2	Straw	Paddy	25 years	increase in TOC (0.86 g/kg)/MBC ^a (60.3 mg/kg)	Yan et al. (2007)
3	Straw	Paddy	6 months	C sequestration ranged from -165 to -89 g/m ² /season	Iqbal et al. (2009)
4	Straw	Rice paddy	-	SOC increased by 12.8 ± 0.4% and active C fraction by 27.4 ± 1.4%-56.6 ± 1.8%	Liu et al. (2014)
5	Straw	Maize, rice, wheat	-	~85.1–730 kg C sequestered/ha/year	Lu et al. (2009)
6	Rice straw	Rice	4 years	1.68–2.32 g/kg increase in TC	Bhattacharyya et al. (2012)
7	Rice straw	Rice	6 months	SOC increased by 0.10 t/ha	Xionghui et al. (2012)
8	Rice straw	Wheat	5 months	0.09% increase in organic C	Pathak et al. (2006)
9	Rice straw, Maize stalks	Rice	2 years	SOC increased by 9.63%	Witt et al. (2000)
10	Wheat straw	Wheat	10 months	3.2–6.3 g/kg increase in SOC	Duiker and Lal (1999)
11	Wheat residues, rice residues	Rice, wheat	4 years	Wheat residues resulted in greater C sequestration than rice residues	Aulakh et al. (2001)
12	Residues	Corn-soy bean rotation	3 years	TC is 32%–41% greater for no tillage and strip tillage compared to chisel plough	Al Kaysi and Yin, 2005

^a MBC stands for microbial biomass carbon.

mineralization loss of the labile C pool (Al Kaysi and Yin, 2005; Bhattacharyya et al., 2012; Huang et al., 2004; Liu et al., 2014; Xionghui et al., 2012). But, despite practicing conservation tillage, results on GHG emissions following residue incorporation are contradictory. Straw incorporation enhanced CH₄ emission in paddy fields (although this might largely be due to the waterlogged conditions of rice paddy fields) and CO₂ emission in both wetland and upland ecosystems (Bhattacharyya et al., 2012; Liu et al., 2014; Ma et al., 2009; Xionghui et al., 2012). Emission of N₂O occurred in some cases due to the application of additional N along with residues (due to low C to N ratio) and improved microbial activity (Huang et al., 2004; Pathak et al., 2006). Thus, incorporation of crop residues may in some cases enhance C sequestration (see Table 1b) while leading to C loss in some others. Further studies involving long-term experiments using a wide range of soils and environments are needed to fully understand this phenomenon.

3.3. Effects on plant growth and productivity: Impacts of residue removal from soil

Crop residue removal can deplete SOC pools (Blanco-Canqui, 2013); however, it was found to enhance plant growth and crop productivity (e.g., (Blanco-Canqui and Lal, 2007)). In one exceptional case, the yield declined due to immobilization of N by microbial biomass (Lehmann and Rondon, 2006). However, contradictory yield observations have also been reported. Removal of 50–75% corn stover, a potential non-food herbaceous bioenergy feedstock, was found to reduce grain yield by 1.95 Mg ha⁻¹, while complete removal of stover reduced it by 3.32 Mg ha⁻¹ (Blanco-Canqui and Lal, 2007). In contrast, Wortmann et al. (2016) reported 22 and 20% higher mean aboveground N concentrations in biomass and grain yield with > 75% residue removal compared with no residue removal for irrigated no-till corn. Kenney et al. (2015) argued that soil C pool tends to decrease and crop yields tend to increase with an increase in stover removal, suggesting that removal at rates > 50% might be detrimental to soil water and temperature regimes; however, the difference became insignificant after 3 years of removal. According to Johnson et al. (2014), a critical aspect of utilizing corn stover for bio-energy production is to establish sustainable harvest criteria to avoid depletion of SOC and soil erosion. This includes estimating minimum residue return rates to sustain SOC levels, which is useful only if the study is long enough to ascertain the link between the rate of residue returned and resultant changes of SOC. In broad terms, they concluded that keeping around 6 Mg ha⁻¹ yr⁻¹ residue may be a useful generic rate. Nonetheless, empirical data are necessary to calibrate and refine process-based models in order to

generate sustainable harvest rate guidelines for producers, industries, and policy-makers (Johnson et al., 2014).

4. Meta-analysis for the impact of biochar and crop residue on changes in SOC stocks

Meta-analysis was performed on a selected number of studies shown in Table 1a and b to determine the relative impact of biochar and raw crop residue on SOC using global input factors. Literature from field experiments as well as incubation studies were considered where parameters like mean, standard deviation and sample size were reported to extract the complete dataset; the raw mean difference was introduced as an index of effect size, which denotes the difference between the mean of parameter to be tested and the mean of the control. Figs. 4 and 5 present data for the 95% confidence intervals (CI) calculated using R.

The overall effect (n = 12 studies) of biochar (derived from crop residue) increased SOC stock with the average effect size of 3.32 (CIs: 1.47–5.17) (Fig. 4). The impact of biochar produced from what straws was statistically significant (p < 0.05) with increases in SOC for rice and maize trials reported by Zhang et al. (2010, 2011; 2012; 2018). Mukherjee et al. (2014) and Chan et al. (2008) got similar positive results using biochar from oak wood and green waste, with mean effect sizes of 6.00 and 23.37, respectively. Conversely, results from biochar

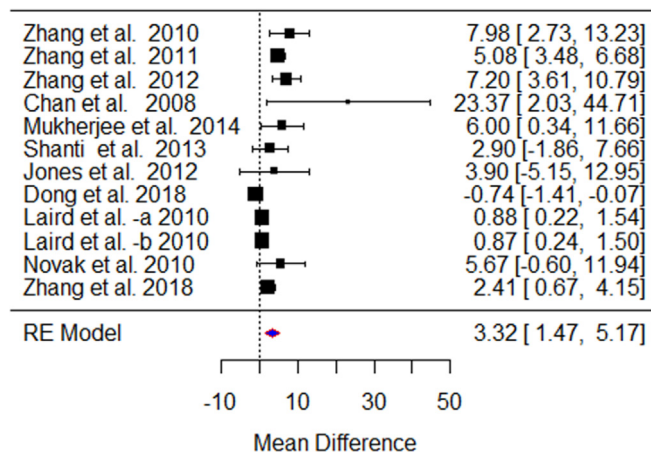


Fig. 4. Forest plot showing the mean effect size of individual observations regarding the status of soil organic carbon with incorporation of biochar amendment. (RE stands for randomized effect model).

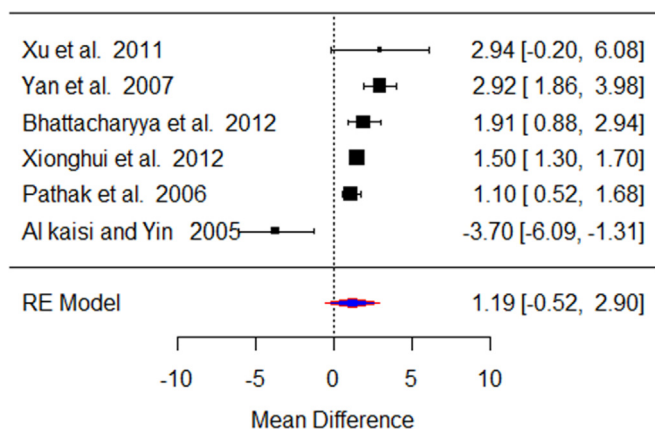


Fig. 5. Forest plot showing the mean effect size of individual observations regarding the status of soil organic carbon with addition of crop residue amendment. (RE stands for randomized effect model).

produced from rubber tree and wood chip was not significant ($p > 0.05$) to the 95% CI, and overlapped the zero effect line (Shanthi et al., 2013; Jones et al., 2012). (Dong et al., 2018) reported a marginal decrease in SOC (mean effect size of -0.74 ; CIs: -1.41 to 0.07) from biochar produced from wheat residue. The incubation studies resulted in non-significant increases in SOC (Laird et al., 2010a,b; Novak et al., 2010).

The impact of raw rice straw as residue significantly increased SOC, but to a lower extent compared to biochar, with mean effective size of 1.19 (CIs: -0.52 – 2.29) (Fig. 5). Total carbon activity marginally increased ($p < 0.05$) during a 10-year study on maize conducted by Xu et al. (2011). SOC was significantly increased by amendment with rice and wheat straw in some of the studies on different crops (Yan et al., 2007; Bhattacharyya et al., 2012; Xionghui et al., 2012; Pathak et al., 2006). Al Kaisi and Yin, 2005 found that soil carbon status significantly decreased ($p < 0.05$) with increased tillage. Their study showed that reducing tillage in a corn-soybean cropping system significantly enhanced the total C content in surface soil compared to strip-tillage, deep rip, chisel plough, and moldboard plough with a concomitant decrease in mean effect size.

5. Effects of biochar and crop residue on soil characteristics

5.1. Physico-chemical properties

Soil amendments such as compost, biochar, and crop residues not only improve soil structure but also enhance soil water retention and soil aggregate formation due to the porous nature and improving saturated hydraulic conductivity (Blanco-Canqui and Lal, 2007; Shanthi et al., 2013). Use of biochar increases pH, CEC (by about 40%), and nutrient retention capacity of soils. Moreover, the large surface area and CEC of biochar leads to reduced pollutant mobility when applied in contaminated soils. The higher CEC of biochar-amended soils is the result of greater surface area and higher charge density per unit surface area (effective surface area) (Lehmann and Rondon, 2006; Liang et al., 2006). As a result, biochar increases soil C storage via organic matter sorption and physical protection. Generally, low temperature biochars increase soil CEC while high temperature biochars raise soil pH (Beesley et al., 2011; Mukherjee et al., 2011; Shanthi et al., 2013; Zhao et al., 2013). Incorporation of crop residues, on the other hand, improves nutrient cycling in the soil (Blanco-Canqui and Lal, 2007). Long-term benefits of nutrient availability include greater stabilization of organic matter, slower nutrient release, and greater retention of cations due to a higher CEC.

5.2. Impact of biochar on soil microbes

Large effective surface area and hydrophobicity of both biochar (depending on pyrolytic temperature) and microorganisms enhances the adhesion of microorganisms to biochar which promotes biological activity in soils (Blackwell et al., 2015; Lehmann and Rondon, 2006; Zimmerman et al., 2011) and stimulates the growth of plant-producing rhizobacteria and fungi (Graber et al., 2010). Biochar is recalcitrant to microbial attack, but it is broken down due to mineralization, which happens because a large part of biochar is mineralized over a short time period while a small portion with a high degree of aromaticity directly contributes to the stable pool of SOC (Lehmann and Rondon, 2006). Biochar can also act as a habitat for extraradical fungal hyphae that sporulate in their micropores due to lower competition from saprophytes, acting as an inoculum for arbuscular mycorrhizal fungi (Lehmann and Rondon, 2006). Biochar also promotes soil micro-organism recovery, illustrated by a study of the effect of biochar derived from rice husk and green waste applied to the soil surface on 1,3-dichloropropene (1,3-D) and chloropicrin (CP) emission and on microbial population re-establishment (Shen et al., 2016).

Biochar can generate volatile organics and polycyclic aromatic hydrocarbons (PAHs) (Dutta et al., 2016b; Quilliam et al., 2012; Spokas et al., 2011b), which can influence soil microbial processes (e.g., nitrification) (Spokas et al., 2010). Both dissolved (Masiello et al., 2013) and volatile (Spokas et al., 2011b) organic C components of biochar are likely to participate in the microbial signaling processes. Recently, Sun et al. (2015) reported that the structure of soil microbial communities is significantly altered by the application of fresh biochar, although such effects could be short-lived. Therefore, the underlying mechanism of some of these observed effects are poorly understood.

6. Mitigation of global warming

The application of crop residue, and biochar produced from them, may offset the emission of GHGs, thereby mitigating global warming.

6.1. Greenhouse gas emission from biochar-treated soils

6.1.1. CH_4

During decomposition, the volatile C fraction in biochar (vs. fixed, defined by Standard, 2013) becomes the predominant source of methanogenic substrates, thereby promoting CH_4 emission (Zhang et al., 2012; Zimmerman, 2010) (Fig. 6). These studies were conducted in paddy fields where waterlogged conditions play an important role in soil CH_4 emission. Methane is an important GHG in waterlogged soil (Liu et al., 2011). However, in some cases, no significant change in the emission of CH_4 was observed with biochar application. This was attributed to the absence of a priming effect which protects the SOM from mineralization by occlusion within biochar pores or sorption of labile organic matter released by biochar onto soil-mineral surfaces (Kuzakov et al., 2000; Mukherjee et al., 2014; Zhang et al., 2012). It is noteworthy that both positive and negative priming effects have been observed in biochar-amended soils, depending on: biochar characteristics, soil type, and time of observation (Zhang et al., 2011). Biochar application also reduces the emission of CH_4 by significantly increasing the abundance of methanotrophic Proteobacteria (Feng et al., 2012; Liu et al., 2011). In addition, post biochar application that leads to CH_4 oxidation can improve soil aeration status (Karhu et al., 2011). The treated soil acts as a sink for atmospheric CH_4 depending upon the soil moisture, diurnal effects of root respiration, or other microbial soil processes that affect the availability of oxygen in the soils (Scheer et al., 2011).

6.1.2. CO_2

Global Warming Potential (GWP) refers to a relative measure of the amount of heat trapped in the atmosphere by GHG relative to CO_2 ;

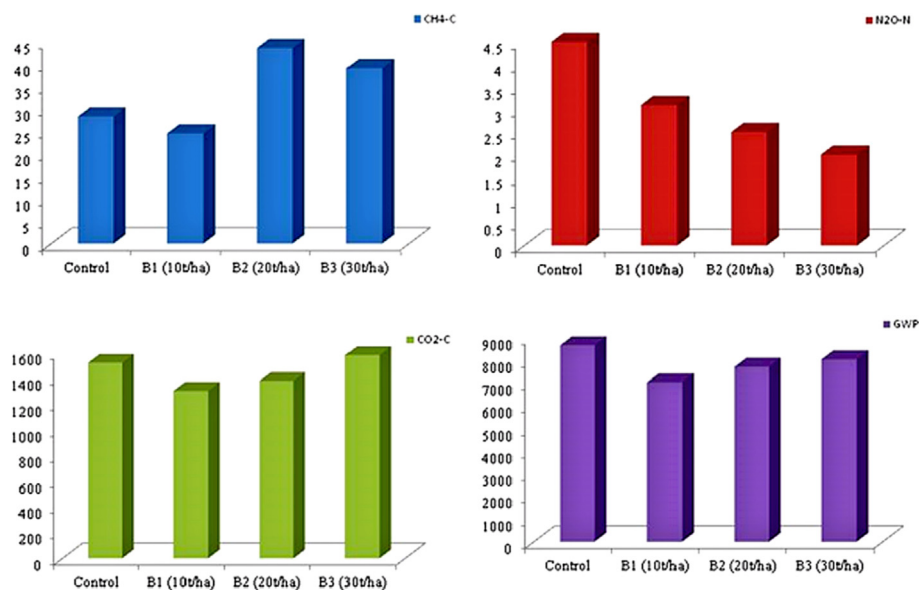


Fig. 6. Emissions of CH₄, N₂O, and CO₂ and their total Global Warming Potential (GWP) in kg/ha at three different application rates (10, 20, and 30 T ha⁻¹) of biochar (source: Zhang et al., 2012).

reports on the impact of biochar are not in agreement. ¹⁴C labeling is used to determine mineralization loss in biochar (Kuzaykov et al., 2009), and has shown that CO₂ can originate from carbonate formed during pyrolysis. Amendment of soil using biochar may sometimes accelerate the decomposition of SOC through enhanced microbial activity, increasing CO₂ emission and total (GWP) (Fig. 6) (Deng et al., 2016; Zhang et al., 2012). Nguyen et al. (2016) reported reduced CO₂ emission from two of three sub-tropical soils on a maize plantation following rice husk biochar application. Ethylene, which is frequently present in biochar, can sometimes inhibit this process (Mukherjee et al., 2014; Spokas et al., 2010; Zhang et al., 2010, 2011, 2012). Biochar application results in both positive (C mineralization) and negative (C stabilization) priming effects. Positive priming effects lead to CO₂ emission and are typical from biochar produced at low temperatures (250–400 °C) while negative priming effects are expected from biochar produced at high temperatures (525–650 °C) (Zimmerman et al., 2011). An increase in the emission of CO₂ may also be attributed to the high internal porosity and adsorbing capacity of biochar, which provides habitat for microorganisms and consequently accelerates the decomposition of adsorbed organic compounds (Rogovska et al., 2011). Decreased bulk density of biochar-amended soils also enhances gas exchange and increases the partial pressure of oxygen in the soil pores, thereby increasing the activity of aerobic microorganisms. However, the increased rate of microbial activity also implies enhanced nutrient cycling, which leads to increased net primary production and C fixation through photosynthesis (Rogovska et al., 2011).

6.1.3. N₂O

Biochar application can reduce N₂O emissions from agricultural fields (Mukherjee et al., 2014; Nguyen et al., 2016; Zhang et al., 2010, 2012, 2015). Enhanced N immobilization and decreased denitrification are two of the main mechanisms for reduced N₂O (Wang et al., 2013). The most recent and plausible explanation is that biochar amendment favors the activity of N₂O reductase (that increases with the increase of soil pH) over other enzymes involved in the conversion of nitrate to N₂O, thereby reducing N₂O emission (Mukherjee et al., 2014; Zhang et al., 2010, 2011). Another explanation is that biochar has the capacity to function as an “electron shuttle” facilitating the transfer of electrons to soil denitrifying microorganisms, which promotes the reduction of N₂O to N₂ (Cayuela et al., 2013, 2014). Inhibition of N₂O emission may arguably be driven by some abiotic factors like increased soil aeration

and the sorption of ammonium and nitrate ions on biochar surfaces, as well as the presence of some inhibitory or toxic organic compounds (Mukherjee et al., 2014; Zhang et al., 2011). There are also some exceptions; for example, application of biochar to a calcisol resulted in a 54% increase in soil N₂O emission (Sánchez-García et al., 2014).

6.2. Greenhouse gas emission from crop residue treated soils

6.2.1. CH₄

The incorporation of crop residues causes increased CH₄ emission due to the decomposition of organic matter, which is the predominant source of methanogenic substrate. Furthermore, labile C in some rice cultivars leaks from roots damaged by excessive anaerobic conditions and is metabolized to its equivalent as CH₄. Methane emission, resulting from C mineralization under anaerobic conditions, is further enhanced due to fertilization and availability of C resources for microbes (Bhattacharyya et al., 2012; Liu et al., 2014; Ma et al., 2009; Xionghui et al., 2012). In a study focusing on straw incorporation in a paddy field of China, CH₄ emission was found to increase by a factor of 1.6–10.5; however, this was largely incumbent upon the water regime, drainage system, and tillage practices. Poor drainage and conservation tillage practice considerably reduced the emission (Ma et al., 2009; Zou et al., 2005). Knoblauch et al. (2010) suggested that charring rice and incorporating the residues (carbon black) in lieu of untreated raw residues may reduce CH₄ emissions from paddy fields by 80%. Haefele et al. (2011) are of the opinion that this is dependent upon site-specific conditions and hence needs further long-term investigations to better understand the phenomenon.

6.2.2. CO₂

The emission of CO₂ upon crop residue incorporation occurs due to a rapid increase in microbial decomposition of the labile SOM pool. Low C/N ratio of the residues results in quicker mineralization, which in turn improves microbial as well as plant root respiration leading to increased CO₂ emission (Bhattacharyya et al., 2012; Iqbal et al., 2009). Therefore, emission varies based on the type of crop residue (Fig. 7) and shows negative correlation with the C:N ratio of the applied plant residues (some of these aspects have been discussed in section 3.2). The removal of crop residues from land, on the other hand, can increase the risk of soil erosion, which ultimately leads to a depletion of SOC, further augmenting CO₂ emission (Lal, 2005).

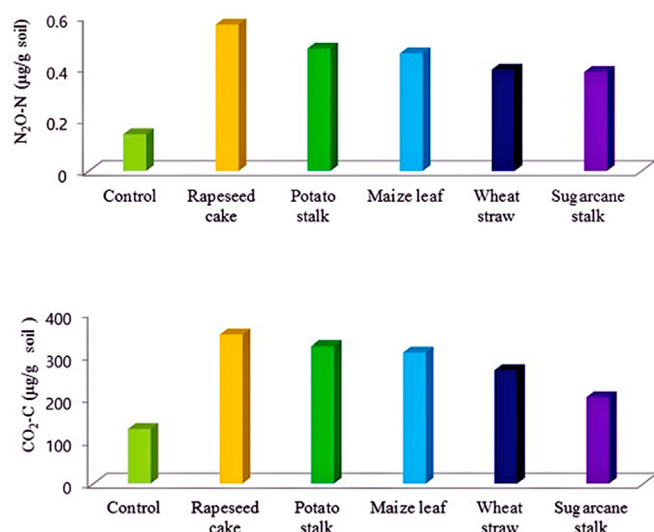


Fig. 7. Emission of N₂O (µg N/g soil) and CO₂ (µg C/g soil) due to the application of different types of crop residues (source: Huang et al., 2004).

6.2.3. N₂O

Nitrous oxide emission resulting from the application of raw residues occurs due to microbial nitrification during aerobic, fallow periods and denitrification when the soils are wetter during anaerobic growing seasons; this is highly dependent on the type of crop residue (Fig. 7) and soil mineral N content (Nguyen et al., 2016; Wu et al., 2016; Zou et al., 2005). Removal of residues, on the other hand, has been shown to enhance N₂O emission under conservation tillage (Congreves et al., 2017). The availability of organic C, nutrients, and soil moisture are major factors for soil denitrification (Dutta et al., 2015, 2016a), which is an anaerobic microbial process involving conversion of nitrate to nitrogen gas. Denitrification depends on the quantity of organic C readily available to the denitrifiers because they are heterotrophs and require organic C for energy. When straw with large C:N ratios is applied, immobilization of fertilizer N occurs, reducing the availability of N (Singh et al., 2005; Wu et al., 2016) and thereby reducing soil N₂O emissions. Moreover, crop residue return may stimulate microbial growth and activity to a great extent and thus create partially anaerobic microsites, resulting in increased soil N₂O production.

7. Economic profitability

Although C sequestration in agricultural systems is a potential win-win strategy, the major challenge for such schemes lies in achieving an equitable participation from farming communities traditionally using cook stoves and other low-tech thermochemical conversion devices (Whitman et al., 2011). Until recently, it was believed that the production cost and pricing of biochar limits its use. Recently, Vochozka et al. (2016) stated that regardless of production cost, which has significantly reduced due to technological advancements, the cost of biochar is still high due to low supply and high demand. A potential annual income of biochar production from wood residue processed per hectare of eucalyptus plantation in Tasmania (Australia) was estimated as \$179k (2014 value), demonstrating potential economic benefits (Wrobel-Tobiszewska et al., 2015). Such economic benefits were also delineated by Field et al. (2013), who showed that biochar production may cater to both environmental and economic benefits. Incorporation of biochar resulted GHG mitigation and led to economic return as well. Dickinson et al. (2015) performed a cost-benefit analysis of biochar technology use to enhance cereal agriculture in two geo-economic regions: North-Western (NW) Europe and Sub-Saharan Africa, in which they calculated the net present value (NPV) of applying biochar by

setting present total cost against total benefits as a function of biochar performance longevity. While the outcome of the analysis was positive for Sub-Saharan Africa as +0.18 to +1.00 t ha⁻¹ yr⁻¹ in terms of cereal grain production, it was not promising for NW Europe (+0.07–0.28 t ha⁻¹ yr⁻¹) due to uncertainty in longevity and performance. A life cycle assessment (LCA) of the energy, climate-change impacts, and biochar economics studied by Roberts et al. (2010) and presented a similar view. They indicated that the net economic income of \$69 MT⁻¹ dry feedstock under biochar induced biomass pyrolysis. Interestingly, this group went a step further showing that biochar production may turn profitable in the long term with greater agronomic understanding, low-cost production (e.g., from waste), and the introduction of an effective C market that includes biochar as a respected method of C offset. The importance of the C offset market was also indicated previously by Galinato et al. (2011). Based on the C offset model, if the market price of biochar is low enough (lower than \$12.14 and \$100.73 MT⁻¹ when the price of carbon offset is \$1 and \$31 MT CO₂, respectively) so that a farmer will earn a profit after applying biochar to the crop field.

Returning the raw crop residues to the soil, offers a seemingly cost-effective solution for small-scale farmers, since the resources needed for their production are cheap and easily available (Chan et al., 2008; Lal, 2004). However, Zhang et al. (2015) contradicted this notion in that no tillage without crop return generated more economic and environmental benefits than crop return. Additionally, there may be a trade-off between the value of crop residue as livestock feed and value for soil protection and productivity (Anderson et al., 2015). Therefore, the relative value of residues for different purposes needs to be elucidated for a range of situations to reach a definitive conclusion. Nonetheless, the efficient use of residues in soil instead of their removal for other competing purposes (i.e. use as fuel, fodder, etc.) should be encouraged.

8. Conclusions and recommendations for future research

Both biochar and crop residues lead to SC gains, albeit with a few exceptions, when increased C losses were observed (e.g., enhanced CH₄ emission observed following the addition of wheat residue). While biochar was found to have dual effects on SC mineralization, depending upon the source material and pyrolysis temperature, the crop residues led to an increase in SOC. Biochar produced from wheat straw-residue were more effective in sequestering C than rice straw-biochars in several studies. However, many pieces of information from fragmented lab experiments using non-commercial biochar, in a very short field experiment or under lab conditions or in soils rich in SOC, could create questions for understanding the potential role of biochar in agriculture (Zhang et al., 2016). Overall, biochar seems to have a greater positive impact on total soil SC while the incorporation of crop residues increases the microbial biomass C of soil. Removal of crop residue, however, has been found to deplete the soil C status in several cases.

In order to achieve higher SC sequestration via soil amendments produced from raw crop residue or biochar, the following management parameters should be considered: tillage method, type of crop residue (C:N), pyrolytic conditions (temp, duration, etc.). Indirect effects of using these amendments should be considered in the context of production/control of soil GHG, environmental health and economic cost benefits of their use. The combined effects of crop residues and biochar have rarely been investigated (with the exception of one recent study by Nguyen et al. (2016). Comparative studies of the impact of crop residues and biochar produced from the same sources and then applied to the same soil at the same time to the same crop as separate trials is required to help understand their relative benefits for all the desired outcomes. Until this happens, it cannot be said with certainty whether the sequestration of SC will be more pronounced when raw crop residues are converted to biochar for soil treatment, or not! Also, studies based on underlying soil processes fundamental to the development of best management practices for long-term soil SC storage are required

under field conditions which are almost impossible to duplicate under laboratory conditions. Since soil degradation is a major problem in developing countries and is often linked to poverty, more detailed case studies are needed to assess the economic feasibility of SC sequestration using crop residues with and without thermolysis.

References

- Abbasi, M.K., Tahir, M.M., Sabir, N., Khurshid, M., 2015. Impact of the addition of different plant residues on nitrogen mineralization-immobilization turnover and carbon content of a soil incubated under laboratory conditions. *Solid Earth* 6 (1), 197.
- Al Kaisi, M.M., Yin, X., 2005. Tillage and crop residue effects on soil carbon and carbon dioxide emission in corn-soybean rotations. *J. Environ. Qual.* 34 (2), 437–445.
- Anderson, E.K., Lee, D., Allen, D.J., Voigt, T.B., 2015. Agronomic factors in the establishment of tetraploid seeded *Miscanthus × giganteus*. *GCB Bioenergy* 7 (5), 1075–1083.
- Aulakh, M.S., Khera, T., Doran, J.W., Bronson, K.F., 2001. Managing crop residue with green manure, urea, and tillage in a rice–wheat rotation. *Soil Sci. Soc. Am. J.* 65, 820–827.
- Baronti, S., Alberti, G., Delle Vedove, G., Di Gennaro, F., Fellet, G., Genesio, L., Miglietta, F., Peressotti, A., Vaccari, F.P., 2010. The biochar option to improve plant yields: first results from some field and pot experiments in Italy. *Ital. J. Agron.* 5 (1), 3–12.
- Beesley, L., Moreno-Jiménez, E., Gomez-Eyles, J.L., Harris, E., Robinson, B., Sizmur, T., 2011. A review of biochars' potential role in the remediation, revegetation and restoration of contaminated soils. *Environ. Pollut.* 159 (12), 3269–3282.
- Bhattacharyya, P., Roy, K., Neogi, S., Adhya, T., Rao, K., Manna, M., 2012. Effects of rice straw and nitrogen fertilization on greenhouse gas emissions and carbon storage in tropical flooded soil planted with rice. *Soil Tillage Res.* 124, 119–130.
- Blackwell, P., Joseph, S., Munroe, P., Anawar, H.M., Storer, P., Gilkes, R.J., Solaiman, Z.M., 2015. Influences of biochar and biochar-mineral complex on mycorrhizal colonisation and nutrition of wheat and sorghum. *Pedosphere* 25 (5), 686–695.
- Blanco-Canqui, H., 2013. Crop residue removal for bioenergy reduces soil carbon pools: how can we offset carbon losses? *Bioenerg. Res.* 6 (1), 358–371.
- Blanco-Canqui, H., Lal, R., 2007. Soil structure and organic carbon relationships following 10 years of wheat straw management in no-till. *Soil Tillage Res.* 95, 240–254.
- Blanco-Canqui, H., Lal, R., 2009. Crop Residue Management and Soil Carbon Dynamics. *Soil Carbon Sequestration and the Greenh. Eff. (soilcarbonseque)*. pp. 291–309.
- Bolan, N.S., Kunhikrishnan, A., Choppala, G., Thangarajan, R., Chung, J., 2012. Stabilization of carbon in composts and biochars in relation to carbon sequestration and soil fertility. *Sci. Total Environ.* 424, 264–270.
- Bruun, E.W., Ambus, P., Eggsgaard, H., Haugaard-Nielsen, H., 2012. Effects of slow and fast pyrolysis biochar on soil C and N turnover dynamics. *Soil Biol. Biochem.* 46, 73–79.
- Cayuela, M.L., Sánchez-Monedero, M.A., Roig, A., Hanley, K., Enders, A., Lehmann, J., 2013. Biochar and denitrification in soils: when, how much and why does biochar reduce N₂O emissions? *Sci. Rep.* 3 (1732).
- Cayuela, M., Van Zwieten, L., Singh, B., Jeffery, S., Roig, A., Sánchez-Monedero, M., 2014. Biochar's role in mitigating soil nitrous oxide emissions: a review and meta-analysis. *Agric. Ecosyst. Environ.* 191, 5–16.
- Cerdà, A., González-Pelayo, Ó., Giménez-Morera, A., Jordán, A., Pereira, P., Novara, A., Brevik, E.C., Prosdoci, M., Mahmoodabadi, N., Keesstra, S., 2016. Use of barley straw residues to avoid high erosion and runoff rates on persimmon plantations in Eastern Spain under low frequency–high magnitude simulated rainfall events. *Soil Res.* 54 (2), 154–165.
- Chan, K., Van Zwieten, L., Meszaros, I., Downie, A., Joseph, S., 2008. Agronomic values of greenwaste biochar as a soil amendment. *Soil Res.* 45 (8), 629–634.
- Chen, S., Rotaru, A.-E., Shrestha, P.M., Malvankar, N.S., Liu, F., Fan, W., Nevin, K.P., Lovley, D.R., 2014. Promoting interspecies electron transfer with biochar. *Sci. Rep.* 4, 1.
- Congreves, K., Brown, S., Németh, D., Dunfield, K., Wagner-Riddle, C., 2017. Differences in field-scale N₂O flux linked to crop residue removal under two tillage systems in cold climates. *GCB Bioenergy* 9, 666–680.
- Coomes, O.T., Miltner, B.C., 2016. Indigenous charcoal and biochar production: potential for soil improvement under shifting cultivation systems. *Land Degrad. Dev.* 28 (3), 811–821.
- Deng, W., Van Zwieten, L., Lin, Z., Liu, X., Sarmah, A.K., Wang, H., 2016. Sugarcane bagasse biochars impact respiration and greenhouse gas emissions from a latosol. *J. Soils Sediments* 1–9.
- Dickinson, D., Balduccio, L., Buysse, J., Ronsse, F., Huylenbroeck, G., Prins, W., 2015. Cost-benefit analysis of using biochar to improve cereals agriculture. *GCB Bioenergy* 7 (4), 850–864.
- Dong, X., Singh, B.P., Li, G., Lin, Q., Zhao, X., 2018. Biochar application constrained native soil organic carbon accumulation from wheat residue inputs in a long-term wheat-maize cropping system. *Agric. Ecosyst. Environ.* 252, 200–207.
- Drake, J.A., Cavignaro, T.R., Cunningham, S.C., Jackson, W.R., Patti, A.F., 2016. Does biochar improve establishment of tree seedlings in saline sodic soils? *Land Degrad. Dev.* 27 (1), 52–59.
- Dutta, T., Stehouwer, R.C., Dell, C.J., 2015. Linking organic carbon, water content, and nitrous oxide emission in a reclaimed coal mine soil. *Land Degrad. Dev.* 26 (6), 620–628.
- Duiker, S.W., Lal, R., 1999. Crop residue and tillage effects on carbon sequestration in a Luvisol in central Ohio. *Soil Tillage Res.* 52, 73–81.
- Dutta, T., Dell, C., Stehouwer, R., 2016a. Nitrous oxide fluxes from mine soil reclaimed with manure-based materials. *Land Degrad. Dev.* 27, 427–437.
- Dutta, T., Kwon, E., Bhattacharya, S., Jeon, B., Deep, A., Uchimiya, M., Kim, K., 2016b. Polyaromatic hydrocarbons (PAHs) and volatile organic compounds (VOCs) in biochar and biochar-amended soil: a review. *GCB Bioenergy*.
- Enders, A., Hanley, K., Whitman, T., Joseph, S., Lehmann, J., 2012. Characterization of biochars to evaluate recalcitrance and agronomic performance. *Bioresour. Technol.* 114, 644–653.
- Feng, Y., Xu, Y., Yu, Y., Xie, Z., Lin, X., 2012. Mechanisms of biochar decreasing methane emission from Chinese paddy soils. *Soil Biol. Biochem.* 46, 80–88.
- Field, J.L., Keske, C.M., Birch, G.L., DeFoort, M.W., Cotrufo, M.F., 2013. Distributed biochar and bioenergy coproduction: a regionally specific case study of environmental benefits and economic impacts. *GCB Bioenergy* 5 (2), 177–191.
- Galinato, S.P., Yoder, J.K., Granatstein, D., 2011. The economic value of biochar in crop production and carbon sequestration. *Energy Pol.* 39 (10), 6344–6350.
- Graber, E.R., Harel, Y.M., Kolton, M., Cytryn, E., Silber, A., David, D.R., Tschansky, L., Borenshtein, M., Elad, Y., 2010. Biochar impact on development and productivity of pepper and tomato grown in fertigated soilless media. *Plant Soil* 337 (1–2), 481–496.
- Haefele, S., Konboon, Y., Wongboon, W., Amarante, S., Maarifat, A., Pfeiffer, E., Knoblauch, C., 2011. Effects and fate of biochar from rice residues in rice-based systems. *Field Crop. Res.* 121 (3), 430–440.
- Hossain, M.K., Strezov, V., Chan, K.Y., Nelson, P.F., 2010. Agronomic properties of wastewater sludge biochar and bioavailability of metals in production of cherry tomato (*Lycopersicon esculentum*). *Chemosphere* 78 (9), 1167–1171.
- Huang, Y., Zou, J., Zheng, X., Wang, Y., Xu, X., 2004. Nitrous oxide emissions as influenced by amendment of plant residues with different C:N ratios. *Soil Biol. Biochem.* 36 (6), 973–981.
- Iqbal, J., Hu, R., Lin, S., Hatano, R., Feng, M., Lu, L., Ahamadou, B., Du, L., 2009. CO₂ emission in a subtropical red paddy soil (Ultisol) as affected by straw and N-fertilizer applications: a case study in Southern China. *Agric. Ecosyst. Environ.* 131 (3), 292–302.
- Jeffery, S., Abalos, D., Spokas, K.A., Verheijen, F.G., 2015. Biochar Effects on Crop Yield. *Biochar for Environmental Management: Science and Technology*. Earthscan Books Ltd, London, pp. 301–325.
- Johnson, J.M., Novak, J.M., Varvel, G.E., Stott, D.E., Osborne, S.L., Karlen, D.L., Lamb, J.A., Baker, J., Adler, P.R., 2014. Crop residue mass needed to maintain soil organic carbon levels: can it be determined? *BioEnergy Res.* 7 (2), 481–490.
- Jones, D., Rousk, J., Edwards-Jones, G., DeLuca, T., Murphy, D., 2012. Biochar-mediated changes in soil quality and plant growth in a three year field trial. *Soil Biol. Biochem.* 45, 113–124.
- Joseph, S., Camps-Arbestain, M., Lin, Y., Munroe, P., Chia, C., Hook, J., Van Zwieten, L., Kimber, S., Cowie, A., Singh, B., 2010. An investigation into the reactions of biochar in soil. *Soil Res.* 48 (7), 501–515.
- Joseph, S., Husson, O., Graber, E.R., Van Zwieten, L., Taherymoosavi, S., Thomas, T., Nielsen, S., Ye, J., Pan, G., Chia, C., 2015. The electrochemical properties of biochars and how they affect soil redox properties and processes. *Agronomy* 5 (3), 322–340.
- Karhu, K., Mattila, T., Bergström, I., Regina, K., 2011. Biochar addition to agricultural soil increased CH₄ uptake and water holding capacity—results from a short-term pilot field study. *Agric. Ecosyst. Environ.* 140 (1), 309–313.
- Kenney, I., Blanco-Canqui, H., Presley, D.R., Rice, C.W., Janssen, K., Olson, B., 2015. Soil and crop response to stover removal from rainfed and irrigated corn. *GCB Bioenergy* 7 (2), 219–230.
- Klüpfel, L., Keilweil, M., Kleber, M., Sander, M., 2014. Redox properties of plant biomass-derived black carbon (biochar). *Environ. Sci. Technol.* 48 (10), 5601–5611.
- Knoblauch, C., Maarifat, A.-A., Pfeiffer, E.-M., Haefele, S.M., 2010. Degradability of black carbon and its impact on trace gas fluxes and carbon turnover in paddy soils. *Soil Biol. Biochem.* 43 (9), 1768–1778.
- Kuzyakov, Y., Friedel, J., Stahr, K., 2000. Review of mechanisms and quantification of priming effects. *Soil Biol. Biochem.* 32 (11), 1485–1498.
- Kuzyakov, Y., Subbotina, I., Chen, H., Bogomolova, I., Xu, X., 2009. Black carbon decomposition and incorporation into soil microbial biomass estimated by 14 C labeling. *Soil Biol. Biochem.* 41 (2), 210–219.
- Laird, D.A., Fleming, P., Davis, D.D., Horton, R., Wang, B., Karlen, D.L., 2019a. Impact of biochar amendments on the quality of a typical Midwestern agricultural soil. *Geoderma* 158, 443–449.
- Laird, D., Fleming, P., Wang, B., Horton, R., Karlen, D., 2010b. Biochar impact on nutrient leaching from a Midwestern agricultural soil. *Geoderma* 158, 436–442.
- Lal, R., 2004. Soil carbon sequestration to mitigate climate change. *Geoderma* 123 (1), 1–22.
- Lal, R., 2005. World crop residues production and implications of its use as a biofuel. *Environ. Int.* 31 (4), 575–584.
- Lal, R., 2006. Enhancing crop yields in the developing countries through restoration of the soil organic carbon pool in agricultural lands. *Land Degrad. Dev.* 17 (2), 197–209.
- Lal, R., 2008. Carbon sequestration in soil. In *CAB reviews perspectives in agriculture*. Vet. Sci. Nutr. Nat. Res. 3 030.
- Lehmann, J., Joseph, S., 2015. *Biochar for Environmental Management: Science, Technology and Implementation*. Routledge.
- Lehmann, J., Rondon, M., 2006. *Bio-char Soil Management on Highly Weathered Soils in the Humid Tropics. Biological Approaches to Sustainable Soil Systems*. CRC Press, Boca Raton, FL, pp. 517–530.
- Liang, B., Lehmann, J., Solomon, D., Kinyangi, J., Grossman, J., O'Neill, B., Skjemstad, J., Thies, J., Luizao, F., Petersen, J., 2006. Black carbon increases cation exchange capacity in soils. *Soil Sci. Soc. Am. J.* 70 (5), 1719–1730.
- Liang, B., Lehmann, J., Sohi, S.P., Thies, J.E., O'Neill, B., Trujillo, L., Gaunt, J., Solomon, D., Grossman, J., Neves, E.G., 2010. Black carbon affects the cycling of non-black carbon in soil. *Org. Geochem.* 41 (2), 206–213.
- Liu, Y., Yang, M., Wu, Y., Wang, H., Chen, Y., Wu, W., 2011. Reducing CH₄ and CO₂ emissions from waterlogged paddy soil with biochar. *J. Soil Sediment* 11 (6),

- 930–939.
- Liu, C., Lu, M., Cui, J., Li, B., Fang, C., 2014. Effects of straw carbon input on carbon dynamics in agricultural soils: a meta-analysis. *Global Change Biol.* 20 (5), 1366–1381.
- Lovley, D.R., Coates, J.D., Blunt-Harris, E.L., Phillips, E.J., Woodward, J.C., 1996. Humic substances as electron acceptors for microbial respiration. *Nature* 382 (6590), 445–448.
- Lu, F., Wang, X., Han, B., Ouyang, Z., Duan, X., Zheng, H., Miao, H., 2009. Soil carbon sequestrations by nitrogen fertilizer application, straw return and no-tillage in China's cropland. *Global Change Biol.* 15, 281–305.
- Ma, J., Ma, E., Xu, H., Yagi, K., Cai, Z., 2009. Wheat straw management affects CH₄ and N₂O emissions from rice fields. *Soil Biol. Biochem.* 41 (5), 1022–1028.
- Mao, J.-D., Johnson, R., Lehmann, J., Olk, D., Neves, E., Thompson, M., Schmidt-Rohr, K., 2012. Abundant and stable char residues in soils: implications for soil fertility and carbon sequestration. *Environ. Sci. Technol.* 46 (17), 9571–9576.
- Masiello, C.A., Chen, Y., Gao, X., Liu, S., Cheng, H.-Y., Bennett, M.R., Rudgers, J.A., Wagner, D.S., Zygourakis, K., Silberg, J.W., 2013. Biochar and microbial signaling: production conditions determine effects on microbial communication. *Environ. Sci. Technol.* 47 (20), 11496–11503.
- Mukherjee, A., Zimmerman, A., Harris, W., 2011. Surface chemistry variations among a series of laboratory-produced biochars. *Geoderma* 163 (3), 247–255.
- Mukherjee, A., Lal, R., Zimmerman, A., 2014. Effects of biochar and other amendments on the physical properties and greenhouse gas emissions of an artificially degraded soil. *Sci. Total Environ.* 487, 26–36.
- Muñoz, M.A., Guzman, J.G., Zornoza, R., Moreno, F., Faz, A., Lal, R., 2016. Effects of biochar and marble mud on mine waste properties to reclaim tailing ponds. *Land Degrad. Dev.* 27 (4), 1227–1235.
- Nguyen, D.H., Scheer, C., Rowlings, D.W., Grace, P.R., 2016. Rice husk biochar and crop residue amendment in subtropical cropping soils: effect on biomass production, nitrogen use efficiency and greenhouse gas emissions. *Biol. Fertil. Soils* 52 (2), 261–270.
- Nishigaki, T., Shibata, M., Sugihara, S., Mvondo-Ze, A.D., Araki, S., Funakawa, S., 2016. Effect of mulching with vegetative residues on soil water erosion and water balance in an oxisol cropped by cassava in east Cameroon. *Land Degrad. Dev.*
- Novak, J.M., Lima, I., Xing, B., Gaskin, J.W., Steiner, C., Das, K., Ahmedna, M., Rehrah, D., Watts, D.W., Busscher, W.J., 2009. Characterization of designer biochar produced at different temperatures and their effects on a loamy sand. *Ann. Environ. Sci.* 3, 195–206.
- Novak, J.M., Busscher, W.J., Watts, D.W., Laird, D.A., Ahmedna, M.A., Niandou, M.A.S., 2010. Short-term CO₂ mineralization after additions of biochar and switchgrass to a Typic Kandudult. *Geoderma* 154, 281–288.
- Olson, K.R., Al-Kaisi, M.M., Lal, R., Lowery, B., 2014. Experimental consideration, treatments, and methods in determining soil organic carbon sequestration rates. *Soil Sci. Soc. Am. J.* 78 (2), 348–360.
- Pathak, H., Singh, R., Bhatia, A., Jain, N., 2006. Recycling of rice straw to improve wheat yield and soil fertility and reduce atmospheric pollution. *Paddy Water Environ.* 4 (2), 111–117.
- Paustian, K., Lehmann, J., Ogle, S., Reay, D., Robertson, G.P., Smith, P., 2016. Climate-smart soils. *Nature* 532 (7597), 49–57.
- Phukan, I., Ramakrishna, Y., 2014. Prospects of using Biochar as soil amendment in tea. *Two a Bud* 61 (1 & 2).
- Qiao-Hong, Z., Xin-Hua, P., HUANG, T.-Q., Zu-Bin, X., Holden, N., 2014. Effect of biochar addition on maize growth and nitrogen use efficiency in acidic red soils. *Pedosphere* 24 (6), 699–708.
- Quilliam, R.S., Marsden, K.A., Gertler, C., Rousk, J., DeLuca, T.H., Jones, D.L., 2012. Nutrient dynamics, microbial growth and weed emergence in biochar amended soil are influenced by time since application and reapplication rate. *Agric. Ecosyst. Environ.* 158, 192–199.
- Roberts, K.G., Gloy, B.A., Joseph, S., Scott, N.R., Lehmann, J., 2010. Life cycle assessment of biochar systems: estimating the energetic, economic, and climate change potential. *Environ. Sci. Technol.* 44 (2), 827–833.
- Rogovska, N., Laird, D., Cruse, R., Fleming, P., Parkin, T., Meek, D., 2011. Impact of biochar on manure carbon stabilization and greenhouse gas emissions. *Soil Sci. Soc. Am. J.* 75 (3), 871–879.
- Sánchez-García, M., Roig, A., Sánchez-Monedero, M.A., Cayuela, M.L., 2014. Biochar increases soil N₂O emissions produced by nitrification-mediated pathways. *Front. Environ. Sci.*
- Saquin, J., YU, Y., Chiu, P., 2016. Wood-derived black carbon (biochar) as a microbial electron donor and acceptor. *Environ. Sci. Technol. Lett.* 3, 62–66.
- Scheer, C., Grace, P.R., Rowlings, D.W., Kimber, S., Van Zwielen, L., 2011. Effect of biochar amendment on the soil-atmosphere exchange of greenhouse gases from an intensive subtropical pasture in northern New South Wales, Australia. *Plant Soil* 345 (1–2), 47–58.
- Shanthi, P., Renuka, R., Sreekanth, N., Babu, P., Thomas, A., 2013. A study of the fertility and carbon sequestration potential of rice soil with respect to application of biochar and selected amendments. *Ann. Environ. Sci.* 7, 17–30.
- Shen, G., Ashworth, D.J., Gan, J.J., Yates, S.R., 2016. Biochar amendment to the soil surface reduces fumigant emissions and enhances soil microorganism recovery. *Environ. Sci. Technol.* 50, 1182–1189.
- Singh, Y., Singh, B., Timsina, J., 2005. Crop residue management for nutrient cycling and improving soil productivity in rice-based cropping systems in the tropics. *Adv. Agron.* 85, 269–407.
- Skjemstad, J.O., Reicosky, D.C., Wilts, A.R., McGowan, J.A., 2002. Charcoal carbon in US agricultural soils. *Soil Sci. Soc. Am. J.* 66 (4), 1249–1255.
- Smith, P., 2016. Soil carbon sequestration and biochar as negative emission technologies. *Glob. Chang. Biol.*
- Spokas, K.A., Reicosky, D.C., 2009. Impacts of sixteen different biochars on soil greenhouse gas production. *Ann. Environ. Sci.* 3, 179–193.
- Spokas, K.A., Baker, J.M., Reicosky, D.C., 2010. Ethylene: potential key for biochar amendment impacts. *Plant Soil* 333 (1–2), 443–452.
- Spokas, K.A., Cantrell, K.B., Novak, J.M., Archer, D.W., Ippolito, J.A., Collins, H.P., Boateng, A.A., Lima, I.M., Lamb, M.C., McAloon, A.J., 2011a. Biochar: a synthesis of its agronomic impact beyond carbon sequestration. *J. Environ. Qual.* 41 (4), 973–989.
- Spokas, K.A., Novak, J.M., Stewart, C.E., Cantrell, K.B., Uchimiya, M., DuSaire, M.G., Ro, K.S., 2011b. Qualitative analysis of volatile organic compounds on biochar. *Chemosphere* 85 (5), 869–882.
- Sposito, G., 1989. *The Chemistry of Soils*. Oxford University Press.
- Standard, A., 2013. *Standard Test Methods for Proximate Analysis of Coal and Coke by Macro Thermogravimetric Analysis*.
- Stockmann, U., Adams, M.A., Crawford, J.W., Field, D.J., Henakaarchchi, N., Jenkins, M., Minasny, B., McBratney, A.B., de Courcelles, V.d.R., Singh, K., 2013. The knowns, known unknowns and unknowns of sequestration of soil organic carbon. *Agric. Ecosyst. Environ.* 164, 80–99.
- Stout, B., Lal, R., Monger, C., 2016. Carbon capture and sequestration: the roles of agriculture and soils. *Int. J. Agric. Biol. Eng.* 9 (1), 1.
- Sun, D., Meng, J., Liang, H., Yang, E., Huang, Y., Chen, W., Jiang, L., Lan, Y., Zhang, W., Gao, J., 2015. Effect of volatile organic compounds absorbed to fresh biochar on survival of *Bacillus mucilaginosus* and structure of soil microbial communities. *J. Soils Sediments* 15 (2), 271–281.
- Uchimiya, M., Wartelle, L.H., Klasson, K.T., Fortier, C.A., Lima, I.M., 2011. Influence of pyrolysis temperature on biochar property and function as a heavy metal sorbent in soil. *J. Agric. Food Chem.* 59 (6), 2501–2510.
- Vochozka, M., Maroušková, A., Váchal, J., Straková, J., 2016. Biochar pricing hampers biochar farming. *Clean Technol. Environ. Pol.* 1–7.
- Wang, Z., Zheng, H., Luo, Y., Deng, X., Herbert, S., Xing, B., 2013. Characterization and influence of biochars on nitrous oxide emission from agricultural soil. *Environ. Pollut.* 174, 289–296.
- Wegner, B.R., Kumar, S., Osborne, S.L., Schumacher, T.E., Vahyala, I.E., Eynarde, A., 2015. Soil response to corn residue removal and cover crops in Eastern South Dakota. *Soil Sci. Soc. Am. J.* 79 (4), 1179–1187.
- Whitman, T., Nicholson, C.F., Torres, D., Lehmann, J., 2011. Climate change impact of biochar cook stoves in western Kenyan farm households: system dynamics model analysis. *Environ. Sci. Technol.* 45 (8), 3687–3694.
- Witt, C., Cassman, K., Olk, D., Biker, U., Liboon, S., Samson, M., Ottow, J., 2000. Crop rotation and residue management effects on carbon sequestration, nitrogen cycling and productivity of irrigated rice systems. *Plant Soil* 225 (1–2), 263–278.
- Woolf, D., Amonette, J.E., Street-Perrott, F.A., Lehmann, J., Joseph, S., 2010. Sustainable biochar to mitigate global climate change. *Nat. Commun.* 1, 56.
- Wortmann, C.S., Shapiro, C.A., Schmer, M.R., 2016. Residue harvest effects on irrigated, No-till corn yield and nitrogen response. *Agron. J.* 108 (1), 384–390.
- Wrobel-Tobiszewska, A., Boersma, M., Sargison, J., Adams, P., Jarick, S., 2015. An economic analysis of biochar production using residues from Eucalypt plantations. *Biomass Bioenergy* 81, 177–182.
- Wu, Y., Lin, S., Liu, T., Wan, T., Hu, R., 2016. Effect of crop residue returns on N₂O emissions from red soil in China. *Soil Use Manag.* 32, 80–88.
- Xionghui, J., Jiamei, W., Hua, P., Lihong, S., Zhenhua, Z., Zhaobing, L., Faxiang, T., Liangjie, H., Jian, Z., 2012. The effect of rice straw incorporation into paddy soil on carbon sequestration and emissions in the double cropping rice system. *J. Sci. Food Agric.* 92 (5), 1038–1045.
- Xu, M., Lou, Y., Sun, X., Wang, W., Baniyammuddin, M., Zhao, K., 2011. Soil organic carbon active fractions as early indicators for total carbon change under straw incorporation. *Biol. Fert. Soils* 47, 745.
- Xu, W., Pignatello, J.J., Mitch, W.A., 2013. Role of black carbon electrical conductivity in mediating hexahydro-1, 3, 5-trinitro-1, 3, 5-triazine (RDX) transformation on carbon surfaces by sulfides. *Environ. Sci. Technol.* 47 (13), 7129–7136.
- Xu, S., Adhikari, D., Huang, R., Zhang, H., Tang, Y., Roden, E., Yang, Y., 2016. Biochar-facilitated microbial reduction of hematite. *Environ. Sci. Technol.* 50 (5), 2389–2395.
- Yan, D., Wang, D., Yang, L., 2007. Long-term effect of chemical fertilizer, straw, and manure on labile organic matter fractions in a paddy soil. *Biol. Fert. Soils* 44, 93–101.
- Yazdanpanah, N., 2016. CO₂ emission and structural characteristics of two calcareous soils amended with municipal solid waste and plant residue. *Solid Earth* 7 (1), 105.
- Zhang, A., Cui, L., Pan, G., Li, L., Hussain, Q., Zhang, X., Zheng, J., Crowley, D., 2010. Effect of biochar amendment on yield and methane and nitrous oxide emissions from a rice paddy from Tai Lake plain, China. *Agric. Ecosyst. Environ.* 139 (4), 469–475.
- Zhang, A., Liu, Y., Pan, G., Hussain, Q., Li, L., Zheng, J., Zhang, X., 2011. Effect of biochar amendment on maize yield and greenhouse gas emissions from a soil organic carbon poor calcareous loamy soil from Central China plain. *Plant Soil* 351 (1–2), 263–275.
- Zhang, A., Bian, R., Pan, G., Cui, L., Hussain, Q., Li, L., Zheng, J., Zheng, J., Zhang, X., Han, X., 2012. Effects of biochar amendment on soil quality, crop yield and greenhouse gas emission in a Chinese rice paddy: a field study of 2 consecutive rice growing cycles. *Field Crop. Res.* 127, 153–160.
- Zhang, Z., Guo, L., Liu, T., Li, C., Cao, C., 2015. Effects of tillage practices and straw returning methods on greenhouse gas emissions and net ecosystem economic budget in rice-wheat cropping systems in central China. *Atmos. Environ.* 122, 636–644.
- Zhang, D., Pan, G., Wu, G., Kibue, G.W., Li, L., Zhang, X., Zheng, J., Zheng, J., Cheng, K., Joseph, S., 2016. Biochar helps enhance maize productivity and reduce greenhouse gas emissions under balanced fertilization in a rainfed low fertility inceptisol. *Chemosphere* 142, 106–113.
- Zhang, X., Chen, C., Chen, X., Tao, P., Jin, Z., Han, Z., 2018. Persistent effects of biochar on soil organic carbon mineralization and resistant carbon pool in upland red soil, China. *Environ. Earth Sci.* 77, 177.

- Zhao, L., Cao, X., Mašek, O., Zimmerman, A., 2013. Heterogeneity of biochar properties as a function of feedstock sources and production temperatures. *J. Hazard Mater.* 256, 1–9.
- Zimmerman, A.R., 2010. Abiotic and microbial oxidation of laboratory-produced black carbon (biochar). *Environ. Sci. Technol.* 44 (4), 1295–1301.
- Zimmerman, A.R., Gao, B., Ahn, M.-Y., 2011. Positive and negative carbon mineralization priming effects among a variety of biochar-amended soils. *Soil Biol. Biochem.* 43 (6), 1169–1179.
- Zou, J., Huang, Y., Jiang, J., Zheng, X., Sass, R.L., 2005. A 3-year field measurement of methane and nitrous oxide emissions from rice paddies in China: effects of water regime, crop residue, and fertilizer application. *Glob. Biogeochem. Cycles* 19 (2).